Worksheet & Answer Key

The resistance of an electrical conductor is influenced by several factors. The most significant one is the type of material, which can be described as _____ or insulating. For instance, copper is a good _____ while glass is a poor one. The length and cross-sectional area of the conductor also affect its resistance, with longer conductors having higher resistances due to increased _____. Temperature changes can cause the resistance to fluctuate, making it more _____ in cold conditions. Finally, impurities or defects in the material can lead to increased _____, making it less suitable for use.

Word Bank:

insulating, good, increased, more, less, poor, copper, glass

Fill in the blanks:

1. _____ of an electrical conductor is influenced by several factors.
2. The most significant one is the type of material, which can be described as _____ or insulating.
3. Copper is a good _____ while glass is a poor one.
4. Longer conductors having higher resistances due to increased _____.
5. Temperature changes can cause the resistance to fluctuate, making it more _____ in cold conditions.
6. Impurities or defects in the material can lead to increased _____, making it less suitable for use.

■ Answer Key (for teachers only):

1. insulating
2. good
3. increased
4. more
5. less
6. poor
7. copper
8. glass